



Ismail Ifakir

PhD Student



Profile

I am a 25-year-old PhD student passionate about data science. I'm eager to apply my expertise in data science, machine learning, and AI to impactful projects. Driven by curiosity and a commitment to professional growth, I am motivated to contribute to innovative solutions and advance my skills in cutting-edge technologies.



Education

2024 –
Present

Faculty of Sciences Dhar El Mahraz, Fez, Morocco

—— Ph.D student (Multimodal Aspect-Based Sentiment Analysis)

2024

Polydisciplinary Faculty of Nador, Morocco

—— Master's in Data Sciences and Intelligent Systems

2022

Polydisciplinary Faculty of Ouarzazate, Morocco

—— Bachelor's in Mathematics and Computer Science



Projects

2026

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2019

Dynamic Website for a Private School

- Bachelor's Degree Completion Project: Realization of a Dynamic Website for a Private School. The project involved the use of technologies such as PHP, HTML, CSS, and JavaScript

Speaker recognition

- Deep Learning: CNN

Intrusion detection systems

- Classification of Normal and Attack messages using ML.

Image Classification

- Diabetes detection Using CNN, VGG16, ResNet, and ViT.

Research Laboratory Management Application

- Spring Boot, MySQL, HTML, CSS, and JavaScript

Sentiment Analysis of Moroccan Arabic Dialect

- Master's Degree Completion Project: DrijaBERT for embedding; CNN, LSTM, BiLSTM, and GRU as base classifiers; Logistic Regression as the meta-classifier; and Streamlit for the web application.

Generate Question from CVs

- Internship at 3D Smart Factory: Extracted entities from resumes using Named Entity Recognition (NER) with spaCy and RoBERTa.



Contact



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Teaching

Experi-

Practical Work Instructor in Artificial Intelligence — Higher School of Technology of Fez (EST Fez)

- Delivered 60 hours of practical work (TP) in the module *Artificial Intelligence and Industrial Processes* for the *Process Engineering* program during the 2025–2026 academic year.

Practical Work Instructor in Artificial Intelligence — Faculty of Sciences and Techniques of Al Hoceima (FSTH)

- Delivered 10 hours of practical work (TP) in the module *Artificial Intelligence* for the *Data Engineering and Software Development (IDDL, S6)* program during the 2025–2026 academic year.



Student Supervision

Co-

- Co-supervised one Master SDSI student at the Polydisciplinary Faculty of Nador on the topic: *Multimodal Aspect-Based Sentiment Analysis*, 2024–2025.
- Co-supervised two FSTH IDDL students at the Faculty of Sciences and Techniques of Al Hoceima on the topic: *Sentiment Analysis on Social Networks*, 2024–2025.

- Developed a question generation pipeline based on the extracted entities using Google Gemini.

Multimodal Aspect-Based Sentiment Analysis

- RoBERTa for text embedding, ViT for image embedding, and LSTM as the classifier.

Chatbot using RAG

- Developed a complete RAG (Retrieval-Augmented Generation) pipeline from scratch using LangChain. This included ingesting raw data, preprocessing, creating embeddings, and storing them in a vector database, providing the foundation for powerful RAG applications.
 1. Building a retrieval query pipeline with GROQ.
 2. Integrating a vector database into the pipeline.

Multimodal RAG

- Built a Multimodal RAG pipeline with LangChain using PDF data sources (text + images), leveraging OpenAI's multimodal LLM gpt-4.1 and Ollama's multimodal model gemma3:4b.

Agentic AI for Multiple Data Sources

- Developed advanced RAG-based Agentic AI using Ollama LLaMA2, integrating multiple data sources (e.g., Arxiv, Wikipedia, and others) with agents, tools, toolkits, and an agent executor.



Publications

An Attention-Driven Feature Fusion Approach for Multimodal Aspect-Based Sentiment Analysis

- **Journal Paper — Q1**
- Authors: Ismail Ifakir, El Habib Nfaoui, Abderrahim Zannou, and Asmaa Mourhir
- Published in *Big Data and Cognitive Computing* (BDCC), MDPI.
- Proposed ADFF, a three-stage hierarchical attention fusion approach for multimodal aspect-based sentiment analysis.

An Approach Based on Named Entity Recognition and Semantic Analysis for Recruitment Efficiency and Optimization

- **Journal Article — Q3**
- Authors: Ismail Ifakir, Nouredine Mohtaram, El Habib Nfaoui, Abderrahim Zannou, and Mohammed El Hassouni
- Proposed an AI-based recruitment pipeline using NER, transformer-based semantic matching, and LLM-generated interview questions.

Attention-Guided Fusion for Multimodal Aspect-Based Sentiment Analysis

- **Conference Paper**
- Authors: Ismail Ifakir, El Habib Nfaoui, and Abderrahim Zannou
- Proposed a two-stage attention-guided fusion framework for aspect-aware text-image sentiment analysis.

- Co-supervised one FSTH IDDL student at the Faculty of Sciences and Techniques of Al Hoceima on the topic: *Development and Training of an NLP Model for the Classification of Job Offers and Candidate CVs*, 2025–2026.



Skills

- Python,Java,C,JavaScript,HTML/CSS
- Node.js, Flask, Streamlit
- SQL/NOSQL
- ML/DL/NLP/LLMs/Multimodal/RAG
- Generative AI/ Agentic AI
- MongoDB, Cassandra, ETL
- NumPy, Pands, Matplotlib, NLTK
- Scikit-learn,TensorFlow,Keras,PyTorch



Certifications

- **Machine Learning**
- **Deep Learning**
- **Natural Language Processing**
- **Transformer Models**
- **Generative AI**
- **JSIIA 2025 Certificate**
Communication certificate for presenting: *Sentiment Analysis of Moroccan Arabic Dialect*. Ben M'sik Faculty of Sciences, May 2025.
- **ICOA 2025 Certificate**
Communication certificate for presenting: *Attention-Guided Fusion for Multimodal Aspect-Based Sentiment Analysis*. Kenitra, Morocco, October 2025.



Languages

Tamazight	Native
Arabic	Native
English	Advanced
French	Intermediate